

Exercise J - A

1. Let S be the area of the region enclosed by $y = e^{-x^2}$, $y = 0$, $x = 0$ and $x = 1$. Then : [JEE-2012]

यदि S वक्रों $y = e^{-x^2}$, $y = 0$, $x = 0$ तथा $x = 1$ के मध्य घिरा हुआ क्षेत्रफल हो तो : [JEE-2012]

- (A) $S \geq \frac{1}{e}$ (B) $S \geq 1 - \frac{1}{e}$ (C) $S \leq \frac{1}{4} \left(1 + \frac{1}{\sqrt{e}} \right)$ (D) $S \leq \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{e}} \left(1 - \frac{1}{\sqrt{2}} \right)$

Ans. ABD

Sol. Area (OABC) = 1

Shaded area is S.

Clearly $S < 1$

$$\text{and } \int_0^1 e^{-x^2} dx > \int_0^1 e^{-x} dx$$

$$\Rightarrow S > 1 - \frac{1}{e} \quad (\therefore \text{ (B) is correct})$$

Again $S \geq \text{Area (trapezium ACDO)}$

$$\Rightarrow S \geq \frac{1}{2} \left(1 + \frac{1}{\sqrt{e}} \right) \left(\frac{1}{\sqrt{2}} \right)$$

$$\Rightarrow S \geq \frac{1}{2\sqrt{2}} \left(1 + \frac{1}{\sqrt{e}} \right)$$

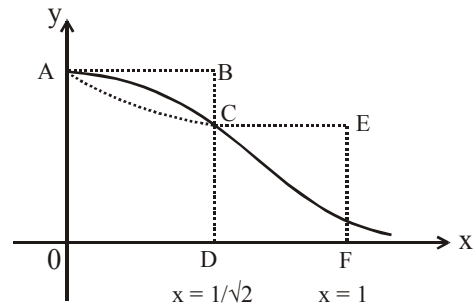
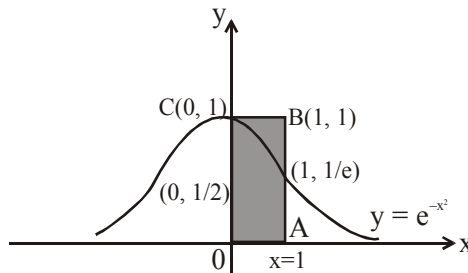
$\therefore$ C is wrong

Also $S \leq \text{Sum of areas of rectangles ABDO \& CEFD}$

$$\Rightarrow S \leq \frac{1}{\sqrt{2}} \times 1 + \left(1 - \frac{1}{\sqrt{2}} \right) \left(\frac{1}{\sqrt{e}} \right)$$

$$\Rightarrow S \leq \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{e}} \left(1 - \frac{1}{\sqrt{2}} \right)$$

$(\therefore \text{ (D) is correct})$



2. The area enclosed by the curves $y = \sin x + \cos x$ and $y = |\cos x - \sin x|$ over the interval $\left[0, \frac{\pi}{2} \right]$ is :

[JEE(Adv.)2013]

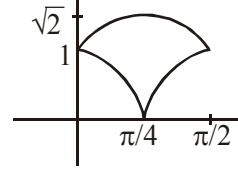
अन्तराल $\left[0, \frac{\pi}{2}\right]$ पर वक्रों $y = \sin x + \cos x$ तथा $y = |\cos x - \sin x|$ द्वारा परिबद्ध क्षेत्रफल है : [JEE(Adv.)2013]

- (A) $4(\sqrt{2}-1)$ (B) $2\sqrt{2}(\sqrt{2}-1)$ (C) $2(\sqrt{2}+1)$ (D) $2\sqrt{2}(\sqrt{2}+1)$

Ans. B

Sol. $y = \sin x + \cos x = \sqrt{2} \sin\left(x + \frac{\pi}{4}\right)$

$$y = |\cos x - \sin x| = \sqrt{2} \left(\cos\left(x + \frac{\pi}{4}\right) \right)$$



Area

$$\begin{aligned} &= \int_0^{\pi/4} [(\sin x + \cos x) - (\cos x - \sin x)] dx \\ &\quad + \int_{\pi/4}^{\pi/2} [(\sin x + \cos x) - (\sin x - \cos x)] dx \\ &= \int_0^{\pi/4} 2 \sin x dx + \int_{\pi/4}^{\pi/2} 2 \cos x dx \\ &= [-2 \cos x]_0^{\pi/4} + [2 \sin x]_{\pi/4}^{\pi/2} \\ &= 2\sqrt{2}(\sqrt{2}-1) \end{aligned}$$

3. Let $f : \left[\frac{1}{2}, 1\right] \rightarrow \mathbb{R}$ (the set of all real numbers) be a positive, non-constant and differentiable function such that

$f'(x) < 2f(x)$ and $f\left(\frac{1}{2}\right) = 1$. Then the value of $\int_{1/2}^1 f(x) dx$ lies in the interval : [JEE(Adv.)2013]

माना कि $f : \left[\frac{1}{2}, 1\right] \rightarrow \mathbb{R}$ (सभी वास्तविक संख्याओं का समुच्चय) एक धनात्मक अचरेतर (non constant) तथा अवकलनीय फलन

है कि $f'(x) < 2f(x)$ तथा $f\left(\frac{1}{2}\right) = 1$ है, तब $\int_{1/2}^1 f(x) dx$ का मान निम्न अन्तराल में है : [JEE(Adv.)2013]

- (A) $(2e - 1, 2e)$ (B) $(e - 1, 2e - 1)$ (C) $\left(\frac{e-1}{2}, e-1\right)$ (D) $\left(0, \frac{e-1}{2}\right)$

Ans. D

Sol. $f'(x) - 2f(x) < 0$

M u l t i p l y b o t h s i d e b y e^{-2x}

$$e^{-2x} f'(x) - 2e^{-2x} f(x) < 0$$

$$\frac{d}{dx}(e^{-2x} f(x)) < 0$$

Now, $g(x) = e^{-2x} f(x)$

$\therefore g(x)$ is a decreasing function.

$$x > \frac{1}{2}$$

$$g(x) < g\left(\frac{1}{2}\right)$$

$$\Rightarrow e^{-2x} f(x) < \frac{1}{e}$$

$$\Rightarrow f(x) < e^{2x-1}$$

$$\Rightarrow \int_{1/2}^1 f(x) dx < \frac{1}{e} \int_{1/2}^1 e^{2x} dx$$

$$= \left[\frac{1}{2e} e^{2x} \right]_{1/2}^1 = \frac{1}{2e} (e^2 - e) = \frac{1}{2} (e - 1)$$

$$\Rightarrow \int_{1/2}^1 f(x) dx < \frac{e-1}{2}$$

obviously $f(x)$ is positive

$$\therefore \int_{1/2}^1 f(x) dx > 0$$

4. Let $F(x) = \int_x^{x^2 + \frac{\pi}{6}} 2 \cos^2 t \, dt$ for all $x \in \mathbb{R}$ and $f: \left[0, \frac{1}{2}\right] \rightarrow [0, \infty)$ be a continuous function. For $a \in \left[0, \frac{1}{2}\right]$, if $F(a)$

+ 2 is the area of the region bounded by $x = 0$, $y = 0$, $y = f(x)$ and $x = a$, then $f(0)$ is. **[JEE(Adv.)2015]**

माना कि सभी $x \in \mathbb{R}$ के लिए, $F(x) = \int_x^{x^2 + \frac{\pi}{6}} 2 \cos^2 t \, dt$ तथा $f: \left[0, \frac{1}{2}\right] \rightarrow [0, \infty)$ एक संतत फलन है। यदि उन सभी $a \in \left[0, \frac{1}{2}\right]$

के लिए $F'(a) + 2$ उस क्षेत्र का क्षेत्रफल है, जो कि $x = 0$, $y = 0$, $y = f(x)$ और $x = a$ से घिरा हुआ है, तब $f(0)$ का मान है।

[JEE(Adv.)2015]

Ans. 3

Sol. $F(x) = \int_x^{x^2 + \frac{\pi}{6}} 2 \cos^2 t \, dt$

$$F'(x) = 2 \left(\cos \left(x^2 + \frac{\pi}{6} \right) \right)^2 2x - 2 \cos^2 x$$

$$\therefore F'(a) + 2 = \int_0^a f(x) dx$$

$$2 \left(\cos \left(a^2 + \frac{\pi}{6} \right) \right)^2 2a - 2 \cos^2 a + 2 = \int_0^a f(x) dx$$

$$\Rightarrow 4 \cos^2 \left(a^2 + \frac{\pi}{6} \right) + 4a \cdot 2 \cos \left(a^2 + \frac{\pi}{6} \right) \cdot \left(-\sin \left(a^2 + \frac{\pi}{6} \right) \right) \times 2a + 4 \cos a \sin a = f(a)$$

$$\therefore f(0) = 4 \left(\frac{\sqrt{3}}{2} \right)^2 = 3$$

5. Match the following column :

[JEE(Adv.)-2015]

Column I

Column II

(A) In a triangle ΔXYZ , let a , b and c be the lengths of the sides opposite to the angles X , Y and Z , respectively. If $2(a^2 - b^2) = c^2$ and

(P) 1

$\lambda = \frac{\sin(X - Y)}{\sin Z}$, then possible values of n for which $\cos(n\pi\lambda) = 0$ is (are)

(B) In a triangle ΔXYZ , let a , b and c be the lengths of the sides opposite to the angles X , Y and Z , respectively. If $1 + \cos 2X - 2 \cos 2Y = 2 \sin X \sin Y$, then possible value(s) of a/b is (are)

(Q) 2

(C) In $\mathbb{R}^2$, let $\sqrt{3} \hat{i} + \hat{j}$, $\hat{i} + \sqrt{3} \hat{j}$ and $\beta \hat{i} + (1 - \beta) \hat{j}$ be the position vectors of X , Y and Z with respect to the origin O , respectively. If the distance of Z

(R) 3

from the bisector of the acute angle of $\overline{OX}$ with $\overline{OY}$ is $\frac{3}{\sqrt{2}}$, then possible

value(s) of $|\beta|$ is (are)

- (D) Suppose that $F(\alpha)$ denotes the area of the region bounded by $x = 0$, $x = 2$, $y^2 = 4x$ and $y = |\alpha x - 1| + |\alpha x - 2| + \alpha x$, where $\alpha \in \{0, 1\}$. (S) 5

Then the value(s) of $F(\alpha) + \frac{8}{3}\sqrt{2}$, when $\alpha = 0$ and $\alpha = 1$, is (are)

(T) 6

निम्न कॉलम को सुमेहित कीजिए :

[JEE(Adv.)-2015]

Column I

Column II

- (A) माना कि एक त्रिभुज ΔXYZ में कोणों X , Y और Z के सामने की भुजाओं की लम्बाईयाँ (P) 1

क्रमशः a , b और c है। माना कि $2(a^2 - b^2) = c^2$ और $\lambda = \frac{\sin(X - Y)}{\sin Z}$ है। यदि

$\cos(n\pi\lambda) = 0$ तब n के संभव मान है।

- (B) माना कि एक त्रिभुज ΔXYZ में कोणों X , Y और Z के सामने की भुजाओं की लम्बाईयाँ (Q) 2

क्रमशः a , b और c है। यदि $1 + \cos 2X - 2\cos 2Y = 2 \sin X \sin Y$ तब a/b के संभव मान है

- (C) माना कि R^2 में, मूल बिन्दु O के सापेक्ष $\sqrt{3}\hat{i} + \hat{j}$, $\hat{i} + \sqrt{3}\hat{j}$ और $\beta\hat{i} + (1 - \beta)\hat{j}$ क्रमशः (R) 3

r एर और Z के स्थिति सदिश है। यदि $\overline{OX}$ और $\overline{OY}$ के न्यून कोण के समद्विभाजक

से Z की दूरी $\frac{3}{\sqrt{2}}$ हो, तो $|\beta|$ का संभव मान है।

- (D) माना कि $F(\alpha)$ उस क्षेत्र के क्षेत्रफल को दर्शाता है जो $x = 0$, $x = 2$, $y^2 = 4x$ और (S) 5

$\psi = |\alpha x - 1| + |\alpha x - 2| + \alpha x$, से घिरा हुआ है, जहाँ $\alpha \in \{0, 1\}$ है। $\alpha = 0$

और $\alpha = 1$ के लिए $F(\alpha) + \frac{8}{3}\sqrt{2}$ का मान है।

(T) 6

Ans. A -PRS, B -P, C - PQ, D - ST

Sol. (A) $2(a^2 - b^2) = c^2$

$$2(4R^2(\sin^2 x - \sin^2 y)) = 4R^2 \sin^2 z$$

$$2(\sin(x - y)\sin(x + y)) = \sin^2 z$$

$$\frac{\sin(x - y)}{\sin z} = \frac{1}{2} = \lambda$$

$$\cos(n\pi\lambda) = \cos\left(\frac{n\pi}{2}\right) = 0$$

$n = \text{odd integer}$

$$(B) 1 + \cos 2x = 2 \cos 2y = 2 \sin x \sin y$$

$$\sin^2 x + \sin x \sin y - 2 \sin^2 y = 0$$

$$(\sin x - \sin y)(\sin x + 2 \sin y) = 0$$

$$\sin x = \sin y$$

$$a = b$$

$$\frac{a}{b} = 1$$

$$(C) \text{ distance of } z \text{ from bisector of } \overrightarrow{OX} \text{ \& } \overrightarrow{OY} = \frac{3}{\sqrt{2}}$$

$$\left(\beta - \frac{1}{2}\right)^2 + \left(\beta - \frac{1}{2}\right)^2 = \frac{9}{2}$$

$$\beta = 2, -1$$

$$|\beta| = 2, 1$$

$$(D) \text{ when } \alpha = 0$$

$$A = 6 - \int_0^2 2\sqrt{x} dx$$

$$= 6 - \frac{8\sqrt{2}}{3}$$

$$\text{when } \alpha = 1$$

$$A = \int_0^1 (3 - x - 2\sqrt{x}) dx + \int_1^2 (x + 1 - 2\sqrt{x}) dx$$

$$= \left| 3x - \frac{x^2}{2} - \frac{4}{3}x^{\frac{3}{2}} \right|_0^1 + \left| \frac{x^2}{2} + x - \frac{4}{3}x^{\frac{3}{2}} \right|_1^2$$

$$= 5 - \frac{8\sqrt{2}}{3}$$

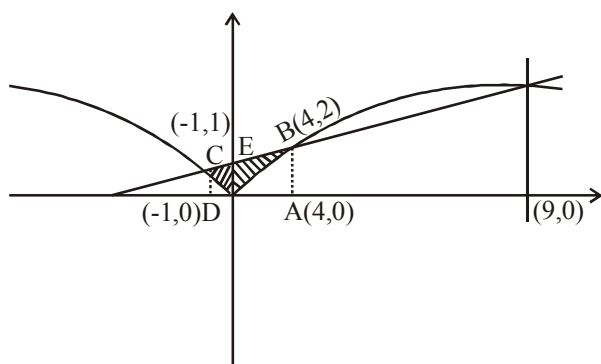
6. Area of the region $\{(x, y) \in \mathbb{R}^2 : y \geq \sqrt{|x+3|}, 5y \leq x+9 \leq 15\}$ is equal to : [JEE(Adv.)2016]

क्षेत्र $\{(x, y) \in \mathbb{R}^2 : y \geq \sqrt{|x+3|}, 5y \leq x+9 \leq 15\}$ का क्षेत्रफल है : [JEE(Adv.)2016]

- (A) $\frac{1}{6}$ (B) $\frac{4}{3}$ (C) $\frac{3}{2}$ (D) $\frac{5}{3}$

Ans. C

Sol. Shifting origin to $(-3, 0)$



$$\text{Area } \{(x, y) \in \mathbb{R}^2 : y \geq \sqrt{|x|}, 5y \leq x+6 \leq 15\}$$

$$\text{Desired Area} = \text{Area of trapezium ABCDA} - [\text{Area DCOD} + \text{Area OBAO}]$$

$$= \frac{1}{2} \times (1+2) \times 5 - \left(\int_{-1}^0 \sqrt{-x} \, dx + \int_0^4 \sqrt{x} \, dx \right)$$

$$= \frac{15}{2} - \left(\frac{2}{3} + \frac{16}{3} \right)$$

$$= \frac{3}{2}$$

7. If the line $x = \alpha$ divides the area of region $R = \{(x, y) \in \mathbb{R}^2 : x^3 \leq y \leq x, 0 \leq x \leq 1\}$ into two equal parts, then : [JEE(Adv.)-2017]

यदि रेखा $x = \alpha$ क्षेत्र $R = \{(x, y) \in \mathbb{R}^2 : x^3 \leq y \leq x, 0 \leq x \leq 1\}$ के क्षेत्रफल को दो बराबर भागों में विभाजित करती है, तब : [JEE(Adv.)-2017]

(A) $0 < \alpha \leq \frac{1}{2}$ (B) $2\alpha^4 - 4\alpha^2 + 1 = 0$ (C) $\alpha^4 + 4\alpha^2 - 1 = 0$ (D) $\frac{1}{2} < \alpha < 1$

Ans. BD

Sol. $y = x^3$

$$\int_0^1 (x - x^3) dx = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$$

$$\int_0^\alpha (x - x^3) dx = \frac{1}{8}$$

$$4\alpha^2 - 2\alpha^4 = 1$$

$$2\alpha^4 - 4\alpha^2 + 1 = 0$$

$$2t^2 - 4t + 1 = 0 \text{ (taking } t = \alpha^2 \text{)}$$

$$t = \frac{4 \pm \sqrt{16 - 8}}{4}$$

$$t = \frac{4 \pm 2\sqrt{2}}{4}$$

$$t = \alpha^2 = 1 \pm \frac{1}{\sqrt{2}}$$

$$\therefore \alpha^2 = 1 - \frac{1}{\sqrt{2}} \Rightarrow \frac{1}{2} < \alpha < 1$$

8. Let $f: [0, \infty) \rightarrow \mathbb{R}$ be a continuous function such that $f(x) = 1 - 2x + \int_0^x e^{x-t} f(t) dt$ for all $x \in [0, \infty)$. Then, which

of the following statement(s) is (are) TRUE ?

[JEE(Adv.)2018]

(A) The curve $y = f(x)$ passes through the point (1, 2)

(B) The curve $y = f(x)$ passes through the point (2, -1)

(C) The area of the region $\{(x, y) \in [0, 1] \times \mathbb{R} : f(x) \leq y \leq \sqrt{1-x^2}\}$ is $\frac{\pi-2}{4}$

(D) The area of the region $\{(x, y) \in [0, 1] \times \mathbb{R} : f(x) \leq y \leq \sqrt{1-x^2}\}$ is $\frac{\pi-1}{4}$

माना कि $f: [0, \infty) \rightarrow \mathbb{R}$ एक ऐसा संतत फलन है कि $f(x) = 1 - 2x + \int_0^x e^{x-t} f(t) dt$ सभी $x \in [0, \infty)$ के लिये। तब निम्नलिखित

में से कौन-कौन से कथन सत्य हैं (हैं)?

[JEE(Adv.)2018]

- (A) वक्र $y = f(x)$ बिन्दु $(1, 2)$ से गुजरता है
 (B) वक्र $y = f(x)$ बिन्दु $(2, -1)$ से गुजरता है
 (C) क्षेत्र $\{(x, y) \in [0, 1] \times \mathbb{R} : f(x) \leq y \leq \sqrt{1-x^2}\}$ का क्षेत्रफल $\frac{\pi-2}{4}$ है
 (D) क्षेत्र $\{(x, y) \in [0, 1] \times \mathbb{R} : f(x) \leq y \leq \sqrt{1-x^2}\}$ का क्षेत्रफल $\frac{\pi-1}{4}$ है

Ans. BC

Sol. $f(x) = 1 - 2x + \int_0^x e^{x-t} f(t) dt \Rightarrow f(x) \cdot e^{-x} = (1 - 2x) \cdot e^{-x} + \int_0^x e^{-t} f(t) dt$

$$\Rightarrow f(x)e^{-x} - e^{-x} \cdot f(x) = -2 \cdot e^{-x} - (1 - 2x) \cdot e^{-x} + e^{-x} \cdot f(x)$$

$$\Rightarrow f(x) - 2f(x) = (2x - 3)$$

$$\therefore f(x) = e^{-2x}$$

$$\therefore y \cdot e^{-x} = \int \left(2x - 3 \right) \cdot e^{-2x} dx$$

$$\Rightarrow y \cdot e^{-2x} = (2x - 3) \cdot \frac{e^{-2x}}{-2} - 2 \int \frac{e^{-2x}}{-2} dx \quad \Rightarrow y \cdot e^{-2x} = -\frac{(2x - 3)e^{-2x}}{2} - \frac{e^{-2x}}{2} + c$$

$$\Rightarrow y \cdot e^{-2x} = -\frac{(2x - 3) - 1}{2} + c \cdot e^{2x} \quad \Rightarrow y = (1 - x) + c \cdot e^{2x}$$

$$\Rightarrow y = (1 - x) + c \cdot e^{2x}$$

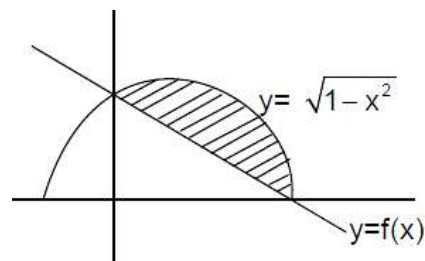
$$\text{put } x = 0$$

$$1 = 1 + c \quad \Rightarrow c = 0$$

$$\therefore y = 1 - x \text{ which passes through point } (2, -1)$$

Now

$$\text{required area} = \frac{1}{4} \cdot \pi \cdot (1)^2 - \frac{1}{2} \cdot 1 \cdot 1 = \frac{\pi}{4} - \frac{1}{2}$$



9. A farmer F_1 has a land in the shape of a triangle with vertices at $P(0,0)$, $Q(1,1)$ and $R(2,0)$. From this land, a neighbouring farmer F_2 takes away the region which lies between the side PQ and a curve of the form $y = x^n$ ($n > 1$). If the area of the region taken away by the farmer F_2 is exactly 30% of the area of $\triangle PQR$, then the value of n is _____. **[JEE(Adv.)2018]**

एक किसान F_1 के पास एक त्रिभुजाकार भूमि है जिसके शीर्ष $P(0,0)$, $Q(1,1)$ तथा $R(2,0)$ पर हैं। एक पड़ोसी किसान F_2 इस भूमि से उस क्षेत्र को ले लेता है जो कि भुजा PQ और $y = x^n$ ($n > 1$) के रूप वाले वक्र के बीच स्थित है। यदि किसान F_2 द्वारा लिये गये क्षेत्र का क्षेत्रफल $\triangle PQR$ के क्षेत्रफल का ठीक 30% है, तब n का मान है। **[JEE(Adv.)2018]**

Ans. 4

Sol. $\frac{30}{100}A = \frac{3}{10}$

$$A = \int_0^1 (x - x^n) dx = \frac{3}{10}$$

$$\left. \frac{x^2}{2} - \frac{x^{n+1}}{n+1} \right|_0^1 = \frac{3}{10}$$

$$\frac{1}{2} - \frac{1}{n+1} = \frac{3}{10}$$

$$n = 4$$

10. The area of the region $\{(x, y): xy \leq 8, 1 \leq y \leq x^2\}$ is :

[JEE(Adv.)2019]

क्षेत्र $\{(x, y): xy \leq 8, 1 \leq y \leq x^2\}$ का क्षेत्रफल है :

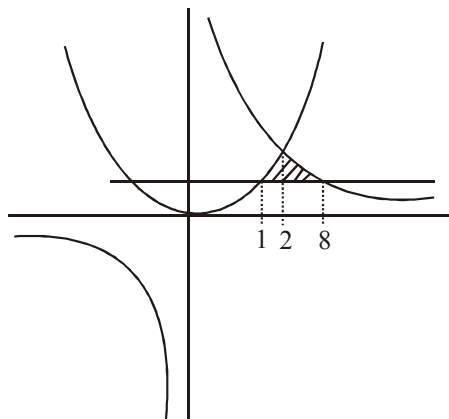
[JEE(Adv.)2019]

(A) $16 \log_e 2 - 6$ (B) $8 \log_e 2 - \frac{14}{3}$ (C) $16 \log_e 2 - \frac{14}{3}$ (D) $8 \log_e 2 - \frac{7}{3}$

Ans. C

Sol. $x^3 = 8$
 $x = 2$

$$\begin{aligned} \text{Area} &= \int_1^2 (x^2 - 1) dx + \int_2^8 \left(\frac{8}{x} - 1 \right) dx \\ &= \left(\frac{x^3}{3} - x \right)_1^2 + (8 \ln x - x)_2^8 \\ &= \left(\frac{8}{3} - 2 - \frac{1}{3} + 1 \right) + [8 \ln 8 - 8 \ln 2 - 8 + 2] \\ &= \left(\frac{7}{3} - 1 \right) + 8 \ln 4 - 6 \\ &= 16 \ln 2 + \frac{7}{3} - 7 \\ &= 16 \ln 2 - \frac{14}{3} \end{aligned}$$



11. Let the functions $f: \mathbb{R} \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = e^{x-1} - e^{-|x-1|}$ and $g(x) = \frac{1}{2}(e^{x-1} + e^{1-x})$. Then the area of the region in the first quadrant bounded by the curves $y=f(x)$, $y=g(x)$ and $x=0$ is :

[JEE(Adv.)2020]

माना कि फलनों $f: \mathbb{R} \rightarrow \mathbb{R}$ एवं $g: \mathbb{R} \rightarrow \mathbb{R}$ को $f(x) = e^{x-1} - e^{-|x-1|}$ एवं $g(x) = \frac{1}{2}(e^{x-1} + e^{1-x})$ के द्वारा परिभाषित किया जाता है। तब प्रथम चतुर्थांश में $y=f(x)$, $y=g(x)$ एवं $x=0$ के द्वारा प्रतिबद्ध क्षेत्र का क्षेत्रफल है?

[JEE(Adv.)2020]

- (A) $(2 - \sqrt{3}) + \frac{1}{2}(e - e^{-1})$ (B) $(2 + \sqrt{3}) + \frac{1}{2}(e - e^{-1})$
(C) $(2 - \sqrt{3}) + \frac{1}{2}(e + e^{-1})$ (D) $(2 + \sqrt{3}) + \frac{1}{2}(e + e^{-1})$

Ans. A

Sol. $f(x) = \begin{cases} e^{x-1} - e^{1-x} & x \geq 1 \\ 0 & x < 1 \end{cases}$

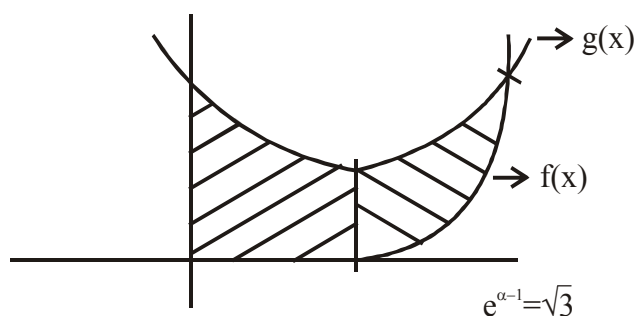
$$g(x) = \frac{1}{2}(e^{x-1} + e^{1-x})$$

$$P\left(1 + \ln \sqrt{3}, \frac{2}{\sqrt{3}}\right) \equiv P\left(\alpha, \frac{2}{\sqrt{3}}\right)$$

$$\text{Required Area} = \int_0^1 \frac{e^{x-1} + e^{1-x}}{2} dx + \int_1^\alpha f(x) - f(x) dx$$

$$= \frac{1}{2} \left(e - \frac{1}{e} \right) + \int_1^\alpha \left(\frac{e^{x-1} + e^{1-x}}{2} - (e^{x-1} - e^{1-x}) \right) dx$$

$$= \frac{1}{2} \left(e - \frac{1}{e} \right) + \int_1^\alpha \left(\frac{3}{2} e^{1-x} - \frac{e^{x-1}}{2} \right) dx = (2 - \sqrt{3}) + \frac{1}{2}(e - e^{-1})$$



12. The area of the region $\left\{ (x, y) : 0 \leq x \leq \frac{9}{4}, 0 \leq y \leq 1, x \geq 3y, x + y \geq 2 \right\}$ is : [JEE(Adv.)2021]

क्षेत्र $\left\{ (x, y) : 0 \leq x \leq \frac{9}{4}, 0 \leq y \leq 1, x \geq 3y, x + y \geq 2 \right\}$ का क्षेत्रफल है : [JEE(Adv.)2021]

(A) $\frac{11}{32}$

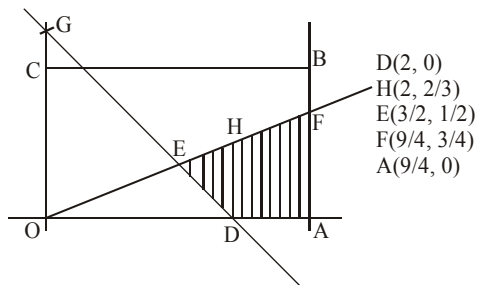
(B) $\frac{35}{96}$

(C) $\frac{37}{96}$

(D) $\frac{13}{32}$

Ans. A

Sol.



$$AB \equiv x = \frac{9}{4}$$

$$BC \equiv y = 1$$

$$DG \equiv x + y = 2$$

$$OF \equiv x = 3y$$

$$DG \equiv x = 2$$

we will divide shaded region in two parts

required area = Area of $\triangle DEH$ + area of trapezium AFHD

$$= \frac{1}{2} \left(\frac{2}{3} \right) \left(\frac{1}{2} \right) + \frac{1}{2} \left(\frac{2}{3} + \frac{3}{4} \right) \left(\frac{1}{4} \right) = \frac{11}{32}$$

13. Consider the functions $f, g: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^2 + \frac{5}{12}$ and $g(x) = \begin{cases} 2 \left(1 - \frac{4|x|}{3} \right) & |x| \leq \frac{3}{4}, \\ 0, & |x| > \frac{3}{4} \end{cases}$. If α is

the area of the region $\left\{ (x, y) \in \mathbb{R} \times \mathbb{R} : |x| \leq \frac{3}{4}, 0 \leq y \leq \min\{f(x), g(x)\} \right\}$ then the value of 9α is ____.

[JEE(Adv.)2022]

फलनों $f, g: \mathbb{R} \rightarrow \mathbb{R}$ पर विचार कीजिए जो $f(x) = x^2 + \frac{5}{12}$ एवं $g(x) = \begin{cases} 2 \left(1 - \frac{4|x|}{3} \right) & |x| \leq \frac{3}{4}, \\ 0, & |x| > \frac{3}{4} \end{cases}$ द्वारा परिभाषित है।

यदि क्षेत्र $\left\{ (x, y) \in \mathbb{R} \times \mathbb{R} : |x| \leq \frac{3}{4}, 0 \leq y \leq \min\{f(x), g(x)\} \right\}$ का क्षेत्रफल α है, तब 9α का मान _____ है।

[JEE(Adv.)2022]

Ans. 6

Sol. $f(x) = x^2 + \frac{5}{12}$

$$g(x) = 2\left(1 - \frac{4}{3}|x|\right)$$

$$|x| \leq \frac{3}{4} \text{ and } 0 \leq y \leq (f(x), g(x))$$

for $x \geq 0$

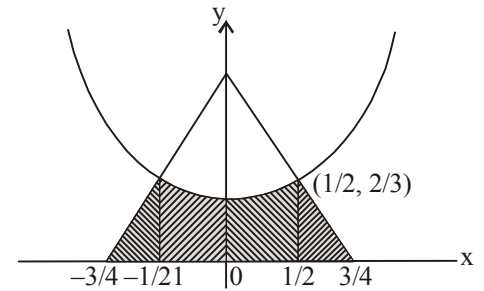
$$x^2 + \frac{5}{12} = 2 - \frac{8}{3}x$$

$$\Rightarrow 12x^2 + 32x - 19 = 0$$

$$12x^2 + 38x - 6x - 19 = 0 \Rightarrow x = \frac{1}{2}$$

$$\text{Required area} = 2\left(\int_0^{1/2} \left(x^2 + \frac{5}{12}\right) dx + \frac{1}{2} \times \frac{1}{4} \times \left(2 - \frac{4}{3}\right)\right)$$

$$\Rightarrow \alpha = 2\left(\frac{1}{24} + \frac{5}{24} + \frac{2}{24}\right) = \frac{2}{3} \text{ sq. units} \Rightarrow 9\alpha = 9 \times \frac{2}{3} = 6$$



14. Let $f: [0, 1] \rightarrow [0, 1]$ be the function defined by $f(x) = \frac{x^3}{3} - x^2 + \frac{5}{9}x + \frac{17}{36}$. Consider the square region $S = [0, 1] \times [0, 1]$. Let $G = \{(x, y) \in S : y > f(x)\}$ be called the green region and $R = \{(x, y) \in S : y < f(x)\}$ be called the red region. Let $L_h = \{(x, h) \in S : x \in [0, 1]\}$ be the horizontal line drawn at a height $h \in [0, 1]$. Then which of the following statements is(are) true? **[JEE(Adv.)2023]**

(A) There exists an $h \in \left[\frac{1}{4}, \frac{2}{3}\right]$ such that the area of the green region above the line L_h equals the area of the green region below the line L_h

(B) There exists an $h \in \left[\frac{1}{4}, \frac{2}{3}\right]$ such that the area of the red region above the line L_h equals the area of the red region below the line L_h

(C) There exists an $h \in \left[\frac{1}{4}, \frac{2}{3}\right]$ such that the area of the green region above the line L_h equals the area of the red region below the line L_h

(D) There exists an $h \in \left[\frac{1}{4}, \frac{2}{3}\right]$ such that the area of the red region above the line L_h equals the area of the green region below the line L_h

माना कि फलन $f : [0,1] \rightarrow [0,1]$, $f(x) = \frac{x^3}{3} - x^2 + \frac{5}{9}x + \frac{17}{36}$ से परिभाषित है। वर्गाकार क्षेत्र (square region)

पर विचार कीजिए। माना कि $G = \{(x, y) \in S : y > f(x)\}$ हरित क्षेत्र (green region) एवं $R = \{(x, y) \in S : y < f(x)\}$ लाल क्षेत्र (red region) कहलाता है। मान लीजिये की $h \in [0,1]$ की ऊँचाई (height) पर खींची गई क्षैतिज रेखा (horizontal line) $L_h = \{(x, h) \in S : x \in [0,1]\}$ है। तब निम्न में से कौनसा (से) कथन सत्य है(हैं)?

[JEE(Adv.)2023]

- (A) एक ऐसा $h \in \left[\frac{1}{4}, \frac{2}{3}\right]$ है कि रेखा L_h के ऊपर के लाल क्षेत्र का क्षेत्रफल रेखा L_h के नीचे के हरित क्षेत्र के क्षेत्रफल के बराबर है
- (B) एक ऐसा $h \in \left[\frac{1}{4}, \frac{2}{3}\right]$ है कि रेखा L_h के ऊपर के हरित क्षेत्र का क्षेत्रफल रेखा L_h के नीचे के लाल क्षेत्र के क्षेत्रफल के बराबर है
- (C) एक ऐसा $h \in \left[\frac{1}{4}, \frac{2}{3}\right]$ है कि रेखा L_h के ऊपर के लाल क्षेत्र का क्षेत्रफल रेखा L_h के नीचे के लाल क्षेत्र के क्षेत्रफल के बराबर है
- (D) एक ऐसा $h \in \left[\frac{1}{4}, \frac{2}{3}\right]$ है कि रेखा L_h के ऊपर के हरित क्षेत्र का क्षेत्रफल रेखा L_h के नीचे के हरित क्षेत्र के क्षेत्रफल के बराबर है

Ans. BCD

Sol. $f(x) = \frac{x^3}{3} - x^2 + \frac{5x}{9} + \frac{17}{36}$

$$f'(x) = x^2 - 2x + \frac{5}{9}$$

$$f'(x) = 0 \text{ at } x = \frac{1}{3} \text{ in } [0,1]$$

A_R = Area of Red region

A_G = Area of Green region

$$A_R = \int_0^1 f(x) dx = \frac{1}{2}$$

Total area = 1

$$\Rightarrow A_G = \frac{1}{2}$$

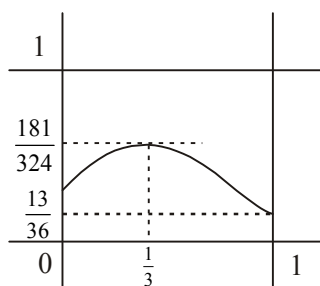
$$\int_0^1 f(x) dx = \frac{1}{2}$$

$$A_G = A_R$$

$$f(0) = \frac{17}{36}$$

$$f(1) = \frac{13}{36} \approx 0.36$$

$$f\left(\frac{1}{3}\right) = \frac{181}{324} \approx 0.558$$



(A) Given statement would be correct when $h = \frac{3}{4}$ but $h \in \left[\frac{1}{4}, \frac{2}{3}\right]$

$\Rightarrow$ (A) is incorrect

(B) Given statement would be correct when $h = \frac{1}{4}$

$\Rightarrow$ (B) is correct

(C) When $h = \frac{181}{324}$, $A_R = \frac{1}{2}$, $A_G < \frac{1}{2}$

$$h = \frac{13}{36}, A_R < \frac{1}{2}, A_G = \frac{1}{2}$$

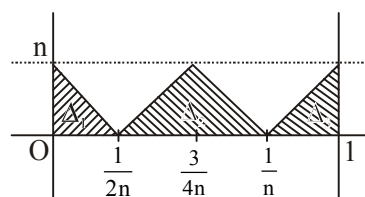
$\Rightarrow A_R = A_G$ for some $h \in \left(\frac{13}{36}, \frac{181}{324}\right)$

$\Rightarrow$ (C) is correct

(D) Option (D) is remaining coloured part of option (C), hence option (D) is also correct.

15. Let $n \geq 2$ be a natural number and $f : [0, 1] \rightarrow \mathbb{R}$ be the function defined by

$$f(x) = \begin{cases} n(1-2nx) & \text{if } 0 \leq x \leq \frac{1}{2n} \\ 2n(2nx-1) & \text{if } \frac{1}{2n} \leq x \leq \frac{3}{4n} \\ 4n(1-nx) & \text{if } \frac{3}{4n} \leq x \leq \frac{1}{n} \\ \frac{n}{n-1}(nx-1) & \text{if } \frac{1}{n} \leq x \leq 1 \end{cases}$$

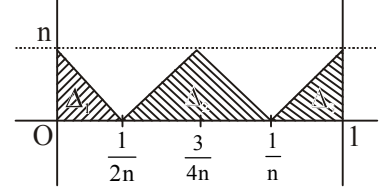


If n is such that the area of the region bounded by the curves $x = 0$, $x = 1$, $y = 0$ and $y = f(x)$ is 4, then the maximum value of the function f is _____.

[JEE(Adv.)2023]

माना कि $n \geq 2$ एक प्राकृत संख्या (natural number) है एवं फलन $f : [0, 1] \rightarrow \mathbb{R}$ इस प्रकार परिभाषित है

$$f(x) = \begin{cases} n(1-2nx) & \text{if } 0 \leq x \leq \frac{1}{2n} \\ 2n(2nx-1) & \text{if } \frac{1}{2n} \leq x \leq \frac{3}{4n} \\ 4n(1-nx) & \text{if } \frac{3}{4n} \leq x \leq \frac{1}{n} \\ \frac{n}{n-1}(nx-1) & \text{if } \frac{1}{n} \leq x \leq 1 \end{cases}$$



यदि n इस प्रकार है कि वक्रों $x = 0$, $x = 1$, $y = 0$ एवं $y = f(x)$ से घिरे क्षेत्र का क्षेत्रफल 4 है तब फलन f का महत्तम मान (maximum value) है _____.

[JEE(Adv.)2023]

Ans. 8

Sol.
$$f(x) = \begin{cases} n(1-2nx) & 0 \leq x \leq \frac{1}{2n} \\ 2n(2nx-1) & \frac{1}{2n} \leq x \leq \frac{3}{4n} \\ 4n(1-nx) & \frac{3}{4n} \leq x \leq \frac{1}{n} \\ \frac{n}{n-1}(nx-1) & \frac{1}{n} \leq x \leq 1 \end{cases}$$

$$\Delta_1 + \Delta_2 + \Delta_3 = 4$$

$$\frac{1}{2}n \times \frac{1}{2n} + \frac{1}{2} \times n \times \frac{1}{2n} + \frac{1}{2} \times n \times \left(1 - \frac{1}{n}\right) = 4$$

$$4 = \frac{n}{2}$$

$$n = 8$$

16. Let $S = \{(x, y) \in \mathbb{R} \times \mathbb{R} : x \geq 0, y \geq 0, y^2 \leq 4x, y^2 \leq 12 - 2x \text{ and } 3y + \sqrt{8}x \leq 5\sqrt{8}\}$ If the area of the region S is $\alpha\sqrt{2}$, then α is equal to :

[JEE(Adv.)2024]

माना कि $S = \{(x, y) \in \mathbb{R} \times \mathbb{R} : x \geq 0, y \geq 0, y^2 \leq 4x, y^2 \leq 12 - 2x \text{ और } 3y + \sqrt{8}x \leq 5\sqrt{8}\}$ है। यदि क्षेत्र (region) S का

क्षेत्रफल $\alpha\sqrt{2}$ है, तब α बराबर है :

[JEE(Adv.)2024]

(A) $\frac{17}{2}$

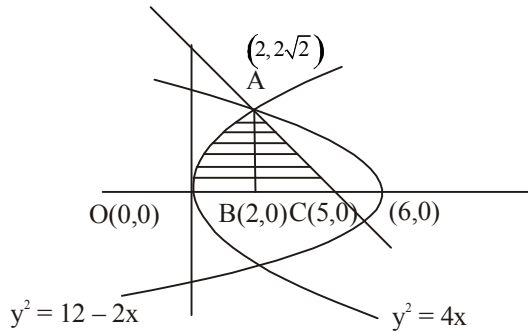
(B) $\frac{17}{3}$

(C) $\frac{17}{4}$

(D) $\frac{17}{5}$

Ans. B

Sol. $3y + \sqrt{8}x = 5\sqrt{8}$



$$4x = 12 - 2x \Rightarrow 6x = 12 \Rightarrow x = 2$$

$$x = 2 \Rightarrow y^2 = 8 \Rightarrow y = \pm 2\sqrt{2}$$

line $3y + \sqrt{8}x = 5\sqrt{8}$ passes through $A(2, 2\sqrt{2})$ & $C(5, 0)$

Required area = Area of OAB + Area of $\triangle ABC$

$$= \int_0^2 2\sqrt{x} dx + \frac{1}{2}(5-2)(2\sqrt{2})$$

$$= \frac{4}{3} \left(\frac{3}{2} x^{\frac{3}{2}} \right) \Big|_0^2 + 3\sqrt{2}$$

$$= \frac{8}{3}\sqrt{2} + 3\sqrt{2}$$

$$= \frac{17}{3}\sqrt{2}$$

CHALLENGER EXERCISES

Exercise C - I

SINGLE CORRECT TYPE QUESTIONS

1. The area bounded by regions $y = \sqrt{x}$ and $x = -\sqrt{y}$ and the curve $x^2 + y^2 = 2$ is : (above x-axis)

- (A) $\frac{\pi}{4}$ (B) $\frac{\pi}{2} - \frac{1}{3}$ (C) $\frac{\pi}{2}$ (D) None

वक्रों $y = \sqrt{x}$, $x = -\sqrt{y}$ तथा $x^2 + y^2 = 2$ से घिरा हुआ क्षेत्रफल होगा : (x-अक्ष के ऊपर)

- (A) $\frac{\pi}{4}$ (B) $\frac{\pi}{2} - \frac{1}{3}$ (C) $\frac{\pi}{2}$ (D) कोई नहीं

Ans. C

Sol. $y = \sqrt{x}$... (1)

$x = -\sqrt{y}$... (2)

$x^2 + y^2 = 2$... (3)

POI of (1) & (3)

$$x^2 + x + 2 = 0$$

$$x = -2, 1$$

In 1st quadrant

$$(1, 1)$$

POI of (2) & (3)

$$y^2 + y - 2 = 0$$

$$y = -2, 1$$

In IInd quadrant

$$y = 1$$

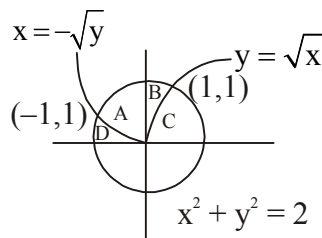
$$x = -1 \quad (-1, 1)$$

Let areas are A, B, C, D

from symmetry

$$A = C \text{ \& } B = D$$

we need



$$= A + B$$

$$= C + D = \text{area of quadrant} = \frac{\pi r^2}{4} = \frac{\pi \cdot (\sqrt{2})^2}{4} = \frac{\pi}{2}$$

2. Area bounded by the curve $y = \max\{\sin x, \cos x\}$ and x-axis, between the lines $x = \frac{\pi}{4}$ and $x = 2\pi$ is equal to :

(A) $\frac{(4\sqrt{2}-1)}{\sqrt{2}}$ sq. units

(B) $(4\sqrt{2}-1)$ sq. units

(C) $\frac{(4\sqrt{2}-1)}{2}$ sq. units

(D) None of these

वक्रों $y = \max\{\sin x, \cos x\}$ तथा x-अक्ष, रेखाओं $x = \frac{\pi}{4}$ व $x = 2\pi$ के मध्य का क्षेत्रफल होगा :

(A) $\frac{(4\sqrt{2}-1)}{\sqrt{2}}$ वर्ग इकाई

(B) $(4\sqrt{2}-1)$ वर्ग इकाई

(C) $\frac{(4\sqrt{2}-1)}{2}$ वर्ग इकाई

(D) इनमें से कोई नहीं

Ans. A

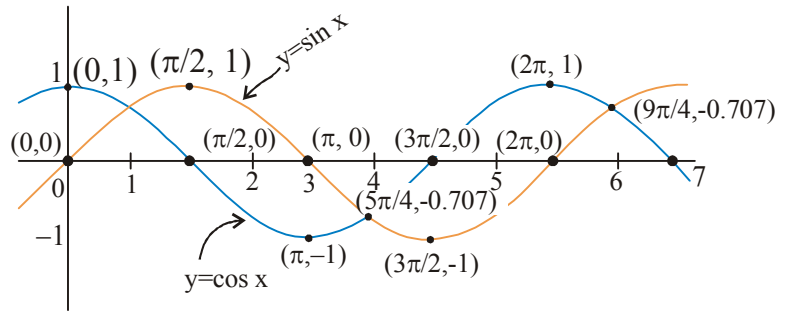
Sol. $y = \max\{\sin x, \cos x\}$, x-axis $x = \frac{\pi}{4}$ & 2π

$$A = \int_{\frac{\pi}{4}}^{\pi} \sin x dx - \int_{\pi}^{\frac{5\pi}{4}} \sin x dx - \int_{\frac{5\pi}{4}}^{\frac{3\pi}{2}} \cos x dx + \int_{\frac{3\pi}{2}}^{2\pi} \cos x dx$$

$$= (-\cos x)_{\frac{\pi}{4}}^{\pi} - [-\cos x]_{\pi}^{\frac{5\pi}{4}} - [\sin x]_{\frac{5\pi}{4}}^{\frac{3\pi}{2}} + [\sin x]_{\frac{3\pi}{2}}^{2\pi}$$

$$= [-\cos x]_{\frac{\pi}{4}}^{\pi} - [-\cos x]_{\pi}^{\frac{5\pi}{4}} - [\sin x]_{\frac{5\pi}{4}}^{\frac{3\pi}{2}} + [\sin x]_{\frac{3\pi}{2}}^{2\pi}$$

$$\begin{aligned}
&= 1 + \frac{1}{\sqrt{2}} + 1 - \frac{1}{\sqrt{2}} + 1 - \frac{1}{\sqrt{2}} + 1 \\
&= 4 - \frac{1}{\sqrt{2}} \\
&= \frac{4\sqrt{2} - 1}{\sqrt{2}}
\end{aligned}$$



3. If $A(n)$ represents the area bounded by the curve $y = n \cdot \ln x$, where $n \in \mathbb{N}$ and $n > 1$, the x -axis and the lines $x = 1$ and $x = e$, then the value of $A(n) + nA(n-1)$ is equal to :

यदि $A(n)$ वक्रों $y = n \cdot \ln x$, (जहाँ $n \in \mathbb{N}$ तथा $n > 1$) x -अक्ष तथा रेखाओं $x = 1$ व $x = e$ के मध्य का क्षेत्रफल हो, तो $A(n) + nA(n-1)$ का मान होगा :

- (A) $\frac{n^2}{e+1}$ (B) $\frac{n^2}{e-1}$ (C) n^2 (D) $e n^2$

Ans. C

Sol. $y = n \ln x$, $n \in \mathbb{N}$, $n > 1$, x -axis

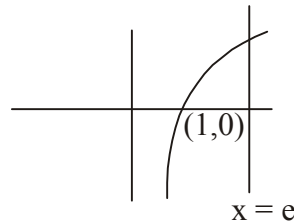
$$A(n) = n \int_1^e \ln x \cdot dx$$

$$= n(x \ln x - x)_1^e = n$$

$$\Rightarrow A(n) = n$$

$$A(n) + nA(n-1) = n + n(n-1)$$

$$= n^2$$



4. Suppose $y = f(x)$ and $y = g(x)$ are two continuous functions whose graphs intersect at the three points $(0, 4)$, $(2, 2)$ and $(4, 0)$ with $f(x) > g(x)$ for $0 < x < 2$ and $f(x) < g(x)$ for $2 < x < 4$. If $\int_0^4 [f(x) - g(x)] dx = 10$ and

$$\int_2^4 [g(x) - f(x)] dx = 5, \text{ the area between two curves for } 0 < x < 2, \text{ is :}$$

माना $y = f(x)$ तथा $y = g(x)$ दो सतत् फलन हैं जिनके वक्र तीन बिन्दुओं $(0, 4)$, $(2, 2)$ तथा $(4, 0)$ पर प्रतिच्छेद होते हैं साथ ही

$f(x) > g(x)$ for $0 < x < 2$ तथा $f(x) < g(x)$ जब $2 < x < 4$. यदि $\int_0^4 [f(x) - g(x)] dx = 10$ तथा $\int_2^4 [g(x) - f(x)] dx = 5$, तो

$\int_0^2 [f(x) - g(x)] dx$ का मान $0 < x < 2$ के लिए होगा :

(A) 5

(B) 10

(C) 15

(D) 20

Ans. C

Sol. Given $\int_0^4 f(x) dx - \int_0^4 g(x) dx = 10$

$$(A_1 + A_3 + A_4) - (A_2 + A_3 + A_4) = 10$$

$$A_1 - A_2 = 10 \quad \dots(1)$$

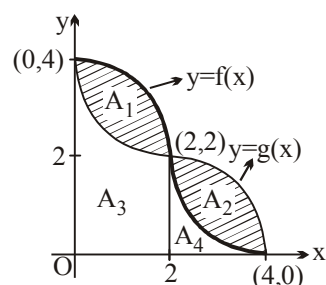
again $\int_2^4 g(x) dx - \int_2^4 f(x) dx = 5$

$$(A_2 + A_4) - A_4 = 5$$

$$A_2 = 5 \quad \dots(2)$$

$$\therefore (1) + (2)$$

$$A_1 = 15$$



5. Let 'a' be a positive constant number. Consider two curves $C_1: y = e^x$, $C_2: y = e^{a-x}$. Let S be the area of the part surrounding by C_1 , C_2 and the y-axis, then $\lim_{a \rightarrow 0} \frac{S}{a^2}$ equals :

'a' एक धनात्मक अचर है। मान लीजिए दो वक्र हैं। $C_1: y = e^x$, $C_2: y = e^{a-x}$ । माना S उस भाग का क्षेत्रफल है जो C_1 , C_2 तथा y-अक्ष

surrounding by C_1 , C_2 and the y-axis, then $\lim_{a \rightarrow 0} \frac{S}{a^2}$ का मान होगा :

(A) 4

(B) 1/2

(C) 0

(D) 1/4

Ans. D**Sol.** Solving $e^x = e^{a-x}$, we get

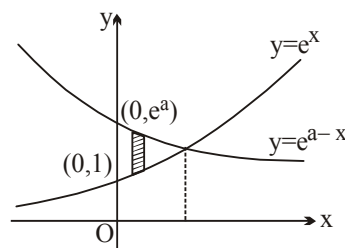
$$e^{2x} = e^a \Rightarrow x = \frac{a}{2}$$

$$S = \int_0^{a/2} (e^a \cdot e^{-x} - e^x) dx = \left[-e^a \cdot e^{-x} + e^x \right]_0^{a/2}$$

$$= (e^a + 1) - (e^{a/2} + e^{a/2}) = e^a - 2e^{a/2} + 1 = (e^{a/2} - 1)^2$$

$$\therefore \frac{S}{a^2} = \left(\frac{e^{a/2} - 1}{a} \right)^2 = \frac{1}{4} \left(\frac{e^{a/2} - 1}{a/2} \right)^2$$

$$\therefore \lim_{a \rightarrow 0} \frac{S}{a^2} = \frac{1}{4}$$



6. The area of the region(s) enclosed by the curves $y = x^2$ and $y = \sqrt{|x|}$ is :

वक्रों $y = x^2$ तथा $y = \sqrt{|x|}$ द्वारा परिबद्ध क्षेत्र (क्षेत्रों) का क्षेत्रफल होगा :

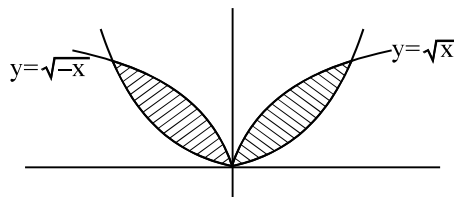
- (A) $\frac{1}{3}$ (B) $\frac{2}{3}$ (C) $\frac{1}{6}$ (D) 1

Ans. B

Sol. $A = \left(\frac{16ab}{3} \right) \cdot 2$

$a = \frac{1}{4}; b = \frac{1}{4}$

$A = \frac{2}{3}$



7. The slope of the tangent to a curve $y = f(x)$ at $(x, f(x))$ is $2x + 1$. If the curve passes through the point $(1, 2)$ then the area of the region bounded by the curve, the x-axis and the line $x = 1$ is :

वक्र $y = f(x)$ के बिन्दु $(x, f(x))$ पर स्पर्श रेखा की प्रवणता $2x + 1$ है। यदि वक्र बिन्दु $(1, 2)$ से गुजरता है तो वक्र का x-अक्ष तथा रेखा $x = 1$ द्वारा परिबद्ध क्षेत्र का क्षेत्रफल होगा :

- (A) $\frac{5}{6}$ (B) $\frac{6}{5}$ (C) $\frac{1}{6}$ (D) 1

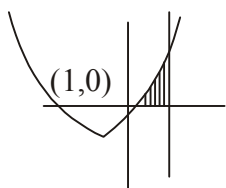
Ans. A

Sol. $\frac{dy}{dx} = 2x + 1$ (1,2)

$y = x^2 + x + c$

$y = x^2 + x$

$y + \frac{1}{4} = \left(x + \frac{1}{2} \right)^2$



$$A = \int_0^1 (x^2 + x) dx$$

$$A = \frac{x^3}{3} + \frac{x^2}{2} \Big|_0^1$$

$$= \frac{1}{3} + \frac{1}{2}$$

$$= \frac{5}{6}$$

8. The area bounded by the curves $y = -\sqrt{-x}$ and $x = -\sqrt{-y}$ where $x, y \leq 0$:

(A) cannot be determined

(B) is $1/3$

(C) is $2/3$

(D) is same as that of the figure bounded by the curves $y = \sqrt{-x}$; $x \leq 0$ and $x = \sqrt{-y}$; $y \leq 0$

वक्रों $y = -\sqrt{-x}$ तथा $x = -\sqrt{-y}$ जहाँ $x, y \leq 0$ द्वारा परिबद्ध क्षेत्रफल होगा :

(A) ज्ञात नहीं किया जा सकता

(B) $1/3$

(C) $2/3$

(D) वही होगा जो वक्रों $y = \sqrt{-x}$; $x \leq 0$ तथा $x = \sqrt{-y}$; $y \leq 0$ द्वारा परिबद्ध है

Ans. B

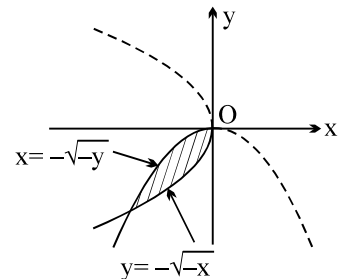
Sol. $y = -\sqrt{-x} \Rightarrow y^2 = -x$ where x & y both $(-)$ ve

$x = -\sqrt{-y} \Rightarrow x^2 = -y$ where x & y both $(-)$ ve

$$\text{Hence } A = \frac{16ab}{3}$$

$$\text{where } a = b = \frac{1}{4}$$

$$\therefore A = \frac{1}{3} \Rightarrow \text{(B)}$$



(A) $\frac{1}{2}$

(B) $\frac{2}{3}$

(C) $\frac{3}{4}$

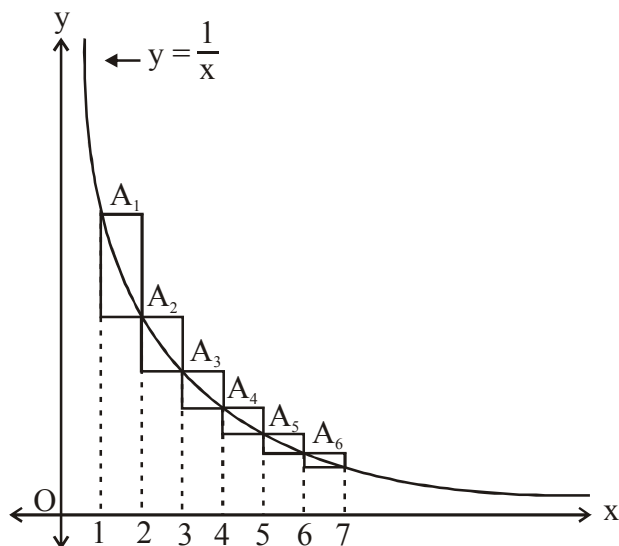
(D) 1

Ans. D

Sol. Method-I:

$$\text{Required sum} = \lim_{n \rightarrow \infty} \left[\left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \dots + \left(\frac{1}{n} - \frac{1}{n+1}\right) \right] = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n+1}\right) = 1$$

Method-II:



$$\text{Area of } r^{\text{th}} \text{ rectangle } A_r = (\text{height}) \times (\text{width}) = \left(\frac{1}{r} - \frac{1}{r+1}\right) \cdot 1$$

$$\begin{aligned} \text{Sum of areas} &= \lim_{n \rightarrow \infty} \sum_{r=1}^n A_r = \lim_{n \rightarrow \infty} \left[\left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \dots + \left(\frac{1}{n} - \frac{1}{n+1}\right) \right] \\ &= \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n+1}\right) = 1 \end{aligned}$$

12. Area enclosed by the figure described by the equation $x^4 + 1 = 2x^2 + y^2$, is :

समीकरण $x^4 + 1 = 2x^2 + y^2$ से घिरे क्षेत्र का क्षेत्रफल होगा :

(A) 3

(B) $\frac{8}{3}$

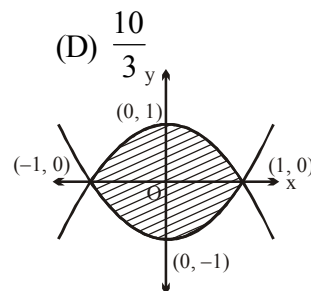
(C) $\frac{7}{3}$

(D) $\frac{10}{3}$

Ans. B

Sol. We have $y^2 = (x^2 - 1)^2 \Rightarrow y = x^2 - 1$ or $y = 1 - x^2$

$$\text{So, required area} = 4 \int_0^1 (1 - x^2) dx = 4 \left[x - \frac{x^3}{3} \right]_0^1 = \frac{8}{3}$$



13. Let $f: [-1, 2] \rightarrow [0, \infty)$ be a continuous function such that $f(x) = f(1-x)$ for all $x \in [-1, 2]$. Let $R_1 = \int_{-1}^2 x f(x) dx$, and R_2 be the area of the region bounded by $y = f(x)$, $x = -1$, $x = 2$, and the x -axis. Then : **[JEE-2011]**

यदि $f: [-1, 2] \rightarrow [0, \infty)$ एक सतत् फलन इस प्रकार है कि $f(x) = f(1-x)$ जहाँ $x \in [-1, 2]$ यदि $R_1 = \int_{-1}^2 x f(x) dx$ तथा R_2

वक्रों $y = f(x)$, $x = -1$, $x = 2$, तथा x -अक्ष से घिरा हुआ क्षेत्रफल है, तो :

[JEE-2011]

- (A) $R_1 = 2R_2$ (B) $R_1 = 3R_2$ (C) $2R_1 = R_2$ (D) $3R_1 = R_2$

Ans. C

Sol. $R_2 = \int_{-1}^2 f(x) dx$, $R_1 = \int_{-1}^2 x f(x) dx$

$$= \int_{-1}^2 (1-x)f(1-x) dx \quad \left(\because \int_a^b f(x) dx = \int_a^b f(a+b-x) dx \right)$$

$$= \int_{-1}^2 (1-x)f(x) dx \quad (\text{given } f(x) = f(1-x))$$

$$= \int_{-1}^2 f(x) dx - \int_{-1}^2 x f(x) dx$$

$$\text{or } R_1 = R_2 - R_1 \Rightarrow 2R_1 = R_2$$